THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $t_n = 2n^2 + 3$, then t_5 is: 1

(a) 53 **(b)** 55 **(c)** 54 **(d)** 52

2. Which term of the sequence $t_n = 3n + 2$ is 53? 1

(a) 16th term **(b)** 18th term **(c)** 17th term **(d)** 19th term

3. A sequence has $t_1 = 4$ and $t_{n+1} = t_n + 3$. Then t_6 equals: 1

(a) 18 **(b)** 19 **(c)** 21 **(d)** 20

4. The 15th term of the AP $5, 8, 11, 14, \ldots$ is: 1

(a) 48 **(b)** 46 **(c)** 49 **(d)** 47

5. The sum of the first 10 terms of the AP $3, 7, 11, 15, \ldots$ is: 1

(a) 210 **(b)** 230 **(c)** 200 **(d)** 220

6. The 5th term of the GP $2, 6, 18, \ldots$ is: 1

(a) 163 **(b)** 164 **(c)** 160 **(d)** 162

7. The number of terms in the AP $3, 7, 11, 15, \ldots$ ending at 83 is: 1

(a) 21 **(b)** 22 **(c)** 23 **(d)** 20

8. A sequence in which each term is obtained by adding a fixed number to the previous term is called: 1

(a) a geometric progression **(b)** a Fibonacci sequence

(c) a constant sequence **(d)** an arithmetic progression

9. If $t_n = 3n^2 - 4$, then t_4 is: 1

(a) 45 **(b)** 44 **(c)** 46 **(d)** 43

10. Which term of the sequence $t_n = 4n - 3$ is 81? 1

(a) 23th term **(b)** 20th term **(c)** 21th term **(d)** 22nd term

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A sequence is defined by $t_n = 3n + 1$. Find t_8 and decide whether 37 is a term. 2

12. Show that $8, 13, 18, 23, \ldots$ is an AP and find its 12th term. 2

13. Write the first five terms of the sequence with $t_1 = 1$ and $t_{n+1} = 2t_n + 1$. 2

14. A sequence is defined by $t_n = 5n - 2$. Find t_7 and decide whether 43 is a term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find which term of the AP $7, 13, 19, 25, \ldots$ equals 151. 3

16. Find the sum of the first 15 terms of the AP $6, 11, 16, 21, \ldots$. 3

17. Find the common ratio of the GP $3, 12, 48, \ldots$ and its 5th term. 3

OR

Find which term of the AP $90, 85, 80, 75, \ldots$ equals 25. 3

18. Find the sum of the first 10 terms of the AP $25, 22, 19, 16, \ldots$. 3

OR

Find the common ratio of the GP $2, 10, 50, \ldots$ and its 4th term. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A taxi charges a fixed ₹200 and ₹40 per km. 5

(i) Show that the fares for 1, 2, 3 km form an AP.

(ii) Find the fare for 12 km.

OR

Rohan saves ₹300 in the first month and ₹50 more each month than in the previous month. 5

(i) How much does he save in month 12?

(ii) How much does he save in 12 months in all?

20. A taxi charges a fixed ₹150 and ₹25 per km. 5

(i) Show that the fares for 1, 2, 3 km form an AP.

(ii) Find the fare for 18 km.

OR

Rohan saves ₹500 in the first month and ₹20 more each month than in the previous month. 5

(i) How much does he save in month 10?

(ii) How much does he save in 10 months in all?

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND